6. Create 3D model of Reading Table and props required on it and set up the Lighting for the same using Unity.

PART 1: Create a New Unity Project

1. Open **Unity Hub**
2. Click **New Project**
3. Choose **3D Core**
4. Name it: `ReadingTableScene`
5. Click **Create**

🎁 PART 2: Set Up the Scene

1. Create Ground (Floor)

- Right-click in **Hierarchy**
- `3D Object → Plane`
- Rename → `Floor`
- Scale it:
  - X = 5, Y = 1, Z = 5

☐ PART 3: Create the Reading Table

1. Table Top

- Right-click → `3D Object → Cube`
- Rename → `TableTop`
- Scale:
  - X = 2
  - Y = 0.1
  - Z = 1
- Position:
  - Y = 1 (height above ground)

---

2. Table Legs (4 legs)

- Right-click → `3D Object → Cube`
- Rename → `Leg1`
- Scale:
  - X = 0.1, Y = 1, Z = 0.1
- Duplicate (Ctrl + D) → make 4 legs
- Position them at corners:
  - Leg1 → (-0.9, 0.5, -0.4)
  - Leg2 → (0.9, 0.5, -0.4)
  - Leg3 → (-0.9, 0.5, 0.4)
  - Leg4 → (0.9, 0.5, 0.4)

📚 PART 4: Add Props on Table

1. Book

- Create Cube → Rename `Book`
- Scale:
  - X = 0.5, Y = 0.05, Z = 0.3
- Place on table:
  - Y ≈ 1.05

---

2. Laptop

Base:

- Cube → `LaptopBase`
- Scale:
  - X = 0.6, Y = 0.05, Z = 0.4

Screen:

- Cube → `LaptopScreen`
- Scale:
  - X = 0.6, Y = 0.4, Z = 0.05
- Rotate:
  - X = -110

---

3. Coffee Cup

- `3D Object → Cylinder`
- Rename → `Cup`

- Scale:
    - X = 0.15, Y = 0.2, Z = 0.15

4. Lamp

Table Lamp Coordinates (Clean Setup)

□ 1. Lamp Base (Cylinder)

- **Position:**
    - X = 0.6
    - Y = 1.05
    - Z = 0
- **Scale:**
    - X = 0.2
    - Y = 0.05
    - Z = 0.2

☞ This sits directly on the table

□ 2. Lamp Stand (Cylinder – thin & tall)

- **Position:**
    - X = 0.6
    - Y = 1.35
    - Z = 0
- **Scale:**
    - X = 0.05
    - Y = 0.3
    - Z = 0.05

☞ Starts from base and goes upward

○ 3. Lamp Head (Sphere)

- **Position:**
    - X = 0.6
    - Y = 1.7
    - Z = 0
- **Scale:**
    - X = 0.25
    - Y = 0.25

    - Z = 0.25

☞ Sits on top of the stand

Important Tip

If alignment looks off:

- Make sure all **X and Z are same (0.6, 0)**
- Only **Y changes** for vertical stacking

🔧 Optional (More Realistic Look)

- Slightly move lamp to corner:
    - X = 0.7, Z = 0.3
- Add **Point Light** inside LampHead:
    - Position same as LampHead
    - Range = 3
    - Intensity = 3

Setup objects manually or use the following coordinates

? Assumptions (important)

- Table center = **(0, 1.05, 0)**
- Table size ≈ **X: 2, Z: 1**
- So usable top area:
    - X: **-1 to +1**
    - Z: **-0.5 to +0.5**

📐 Perfect Top-View Layout (Coordinates)

💻 Laptop (Center Focus)

- **Position:**
    - X = 0
    - Y = 1.05
    - Z = 0
- **Rotation:**
    - Y = 0 (facing forward)

☞ This is your main object → keep it centered

📚 Book (Left Side)

- **Position:**
  - X = -0.5
  - Y = 1.05
  - Z = -0.1
- **Rotation:**
  - Y = 15 (slight tilt for realism)

☞ Looks casually placed, not perfectly straight

☕ Cup (Right Side)

- **Position:**
  - X = 0.5
  - Y = 1.05
  - Z = 0.2
- **Rotation:**
  - Y = 0

☞ Slightly forward so it doesn't clash with laptop

💡 Lamp (Back Corner)

(from previous step, adjusted for layout)

- **Position:**
  - X = 0.7
  - Y = varies (base/stand/head)
  - Z = -0.3

☞ Back-right corner placement (natural study setup)

Spacing Rules (Why this works)

- ✔ Center = main focus (laptop)
- ✔ Left = study element (book)
- ✔ Right = lifestyle element (cup)

- ✓ Back corner = vertical object (lamp)
- ✓ No overlapping → clean rende

PART 5: Add Materials (Colors)

1. Create Material

- Right-click in **Assets**
- `Create → Material`

Create:

- `WoodMaterial` (brown)
- `BookMaterial` (any color)
- `LaptopMaterial` (black/gray)
- `CupMaterial` (white)

☞ Drag materials onto objects

💡 PART 6: Lighting Setup (IMPORTANT)

1. Directional Light (Sun)

- Select **Directional Light**
- Set:
  - Rotation: (50, -30, 0)
  - Intensity: 1–1.5

---

2. Add Point Light (Lamp Light)

- Right-click → `Light → Point Light`
- Rename → `LampLight`
- Position near lamp head

Set:

- Range: 3–5
- Intensity: 2–4
- Color: Warm yellow

---

3. Add Spot Light (Focused Light)

- Right-click → `Light → Spot Light`
- Position above table

Set:

- Angle: 45
- Intensity: ~3

◑ PART 7: Adjust Environment Lighting

- Go to:
  - `Window → Rendering → Lighting`

Change:

- Environment Lighting:
  - Color: Light gray or soft blue

□ PART 8: Shadows

- Select lights
- Enable:
  - Shadows → Soft Shadows

🎥 PART 9: Camera Setup

- Select **Main Camera**
- Position:
  - X = 0, Y = 2, Z = -4
- Rotate:
  - X = 20

✨ PART 10: Final Touches

✓ Add slight rotation to laptop screen
✓ Adjust object spacing neatly
✓ Add multiple books for realism
✓ Slightly vary colors

▶ PART 11: Run the Scene

- Click **Play**
- Check:
  - Lighting effect
  - Object placement
  - Shadows

Add Lamp ON/OFF Script (simple)

```
using UnityEngine;

public class LampToggle : MonoBehaviour
{
    private Light LampLight;

    void Start()
    {
        LampLight = GetComponent<Light>();
    }

    void Update()
    {
        if (Input.GetKeyDown(KeyCode.Space))
        {
            LampLight.enabled = !LampLight.enabled;
        }
    }

}
```

Attach Script & Link Light (Short Steps)

✅ 1. Attach Script

- In **Hierarchy** → click your **Lamp object**
- In **Inspector** → click **Add Component**
- Type → `LampToggle`
- Click it

☞ Script is now attached

✅ 2. Link the Lamp Light

- In **Hierarchy** → find **LampLight (Point Light)**
- Drag it → into the **Lamp Light field** in Inspector (inside LampToggle script)

☞ Done ✅

🎯 Final Check

- Press **Play**
- Press **Spacebar**
  ➡ Light turns ON/OFF

Lamp is the Parent Object

LampHead

LampBase

LampStand are the child objects

Do not attach script to Lamp(Parent Object)

Attach it to LampLight

IF STILL NOT WORKING DO THIS

Switch back to old input system

1. Go to:
   Edit → Project Settings
2. Click **Player**
3. Scroll to **Other Settings**
4. Find **Active Input Handling**
5. Change it to:
   ☞ **Both** *(or "Input Manager (Old)")*
6. Click **Apply**

7. Restart Unity if asked

✓ Now your script will work
✓ Press **Play + Spacebar → Lamp toggles**

Click spacebar to turn lamp ON/OFF